

RiverWatch: Baseline A

Evaluation report | 5 September 2026 | Fictional development fixture

Purpose. Estimate the location of a simulated pipeline leak from pressure measurements. All experiments are simulations; no live pipeline or field pilot is included.

Dataset. There are 600 simulated cases: 420 training, 90 validation and 90 held-out test cases. The final test set is named RW-90. The baseline and candidate reports use the same RW-90 test cases.

Conditions. Water at 20-24 degrees Celsius, one leak per case and complete sensor readings. Sampling frequency is 10 Hz. No artificial measurement noise or sensor outages were introduced.

Exact processor, RAM, GPU and operating-system details were not recorded. Timing was measured on the same bench workstation for both models.

Baseline A: results

Final results on the 90 held-out RW-90 cases:

Measure	Baseline A
Location error (RMSE)	0.84 metres
Mean prediction latency	180 milliseconds
95th-percentile latency	240 milliseconds
Held-out cases	90

RMSE describes aggregate location error. Mean latency and 95th-percentile latency are different measures; they should not be substituted for one another.

No confidence interval, statistical-significance test or per-case error distribution is supplied. These results alone do not establish that a difference is statistically significant.

Baseline A: limitations

No tests of sensor failure, measurement noise, multiple simultaneous leaks or hydrogen service were conducted.

No independent laboratory test, field pilot or operational deployment is reported. The report makes no claim of deployment readiness.

Apply the separate RiverWatch Release Checklist before making a release recommendation. Strong performance on one measure does not replace the other release requirements.